

ELEC2134 : Circuits and Signals

PART A (Week 1 to Week 5)

Transform Methods

- Periodic signals and Fourier series
- Aperiodic signals and Fourier transforms
- Laplace transforms of signals and circuits

System Analysis

- Linear systems and convolution
- Impulse response
- Frequency response and transfer functions
- Steady-state analysis of linear systems
- Transient analysis of 1st and 2nd systems
- Circuit analysis using transform methods

PART B (Week 7 to Week 10)

AC Circuit Analysis

- AC circuits with sinusoidal inputs
- Use of phasors and complex impedance in AC circuit analysis
- AC power: real, reactive and apparent power
- Power factor (leading/lagging)
- Series and parallel resonance
- Transformers and coupled coils
- Application of network theorems: Two port networks

ELEC2134 : Circuits and Signals (PART A)

Topic 4: System Analysis

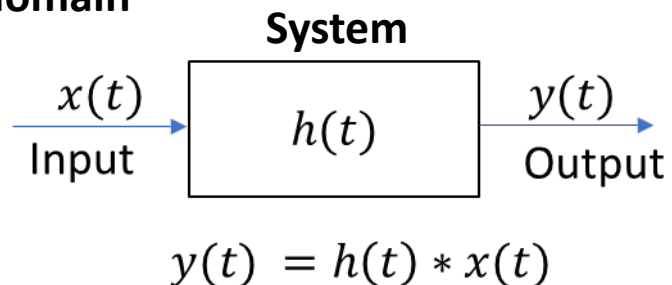
System analysis studies how a system modifies an input signal to produce an output signal. It focuses on the relationship between a system's input and output signals.

Block Diagram Representation of a System

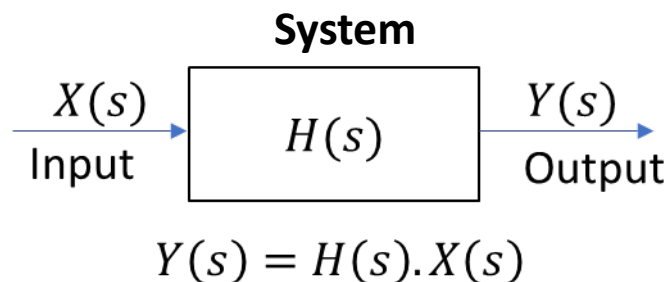
Introduction

- An analogue system is a device (or circuit) that operates on a continuous-time signal $x(t)$, called the input, and produces another continuous-time signal $y(t)$, called the output (or response), according to a well-defined rule.

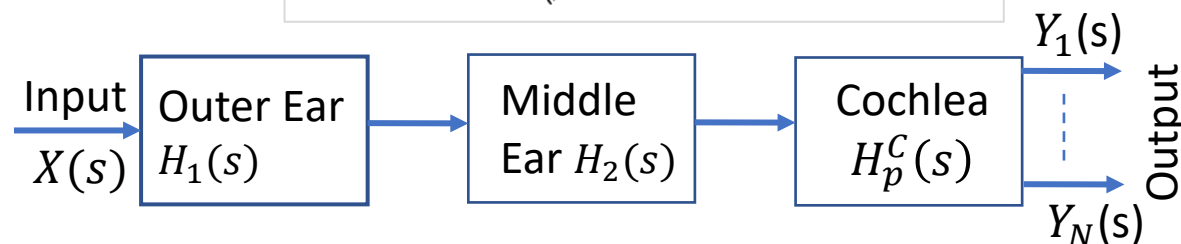
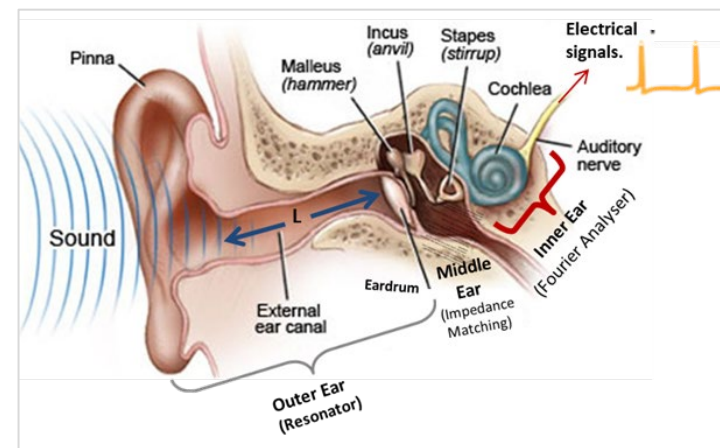
Time domain



Laplace Domain



Peripheral auditory system



It is possible to model the auditory peripheral system comprising of the outer ear transfer function, $H_1(s)$ and the middle ear $H_2(s)$, and the model of the cochlea $H_p^C(s)$ where p goes from 1 to N .

$$Y_1(s) = X(s) H_1(s) H_2(s) H_1^C(s)$$

$$Y_2(s) = X(s) H_1(s) H_2(s) H_2^C(s)$$

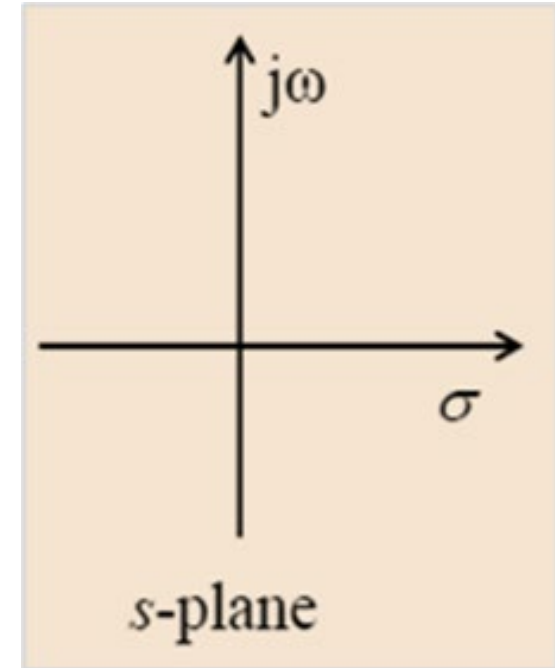
$$Y_N(s) = X(s) H_1(s) H_2(s) H_N^C(s)$$

s-Plane

- It is convenient to represent the complex frequency s ($s = \sigma + j\omega$) graphically in the complex plane, called the s -plane, as shown in Figure 1.
- The horizontal axis represents the real part of s , that is, the exponential damping factor σ , and the vertical axis represents the imaginary part of s , that is, the sinusoidal frequency ω .
- The Fourier Transform can be obtained from the Laplace Transform by setting $\sigma = 0$. Thus, if the region of convergence includes the imaginary axis, then

$$\text{Fourier Transform: } H(j\omega) = H(s) \big|_{s = j\omega}$$

- In the s -plane, $\sigma = 0$ corresponds to the imaginary axis. Therefore, the Fourier transform is obtained by evaluating the Laplace transform along the imaginary axis.
- The $j\omega$ -axis divides the s -plane into two halves. The region of the s -plane to the left of the $j\omega$ -axis is called the left half-plane, while the region to the right of the $j\omega$ -axis is called the right half-plane.



Poles and Zeros

Poles

- When the denominator polynomial $Q(s)$ of a transfer function $\{H(s) = \frac{P(s)}{Q(s)}\}$ of a system is set equal to zero (It is then known as the characteristic equation), the roots of this equation determine the character of the time response of the system.
- The roots of this characteristic equation ($Q(s) = 0$) are called the **poles** of the system.
- For example, if $H(s) = \frac{a}{s+a}$, the pole is at $s = -a$.
- At the poles, the transfer function $H(s)$ becomes infinite.

Zeros

- The roots of the numerator polynomial $P(s)$ of a transfer function $\{H(s) = \frac{P(s)}{Q(s)}\}$ are called the **zeros** of the system.
- For example, if $H(s) = \frac{s+b}{s+a}$, the zero is at $s = -b$.
- At the zeros the function $H(s)$ becomes zero.

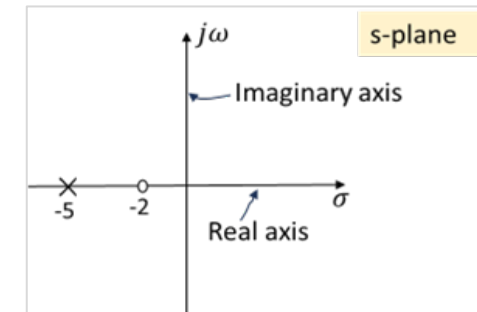
s-plane

- We denote the locations of zeros in the s -plane with the symbol '0' and the locations of poles with the symbol 'x',

Example: If the transfer function of a system

$$\text{is: } H(s) = \frac{Y(s)}{X(s)} = \frac{s+2}{s+5}$$

Plot the pole and zero in the s -plane.

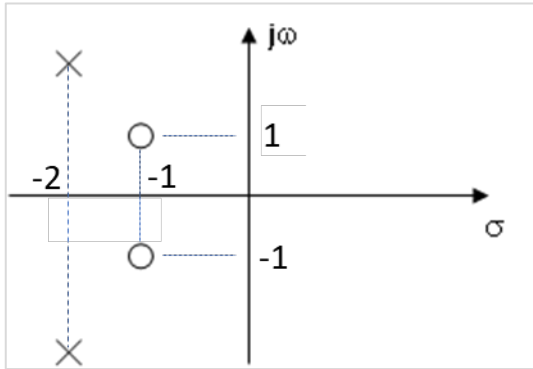


Poles and Zeros

Example: Find the poles and zeros of

$$H(s) = \frac{s^2 + 2s + 2}{s^2 + 4s + 13}$$

- The zeros are obtained by setting the numerator of $H(s)$ to zero:
 $s^2 + 2s + 2 = 0 \Rightarrow s = -1 \pm j$ (complex zeros)
- The poles are obtained by setting the denominator of $H(s)$ to zero:
 $s^2 + 4s + 13 = 0 \Rightarrow s = -2 \pm j3$ (complex poles)



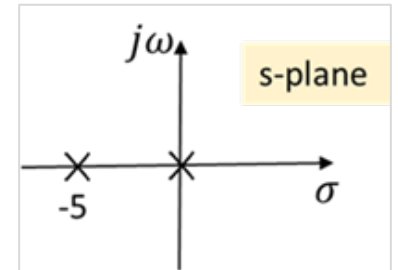
Example: If the transfer function of a system is:

$$H(s) = \frac{Y(s)}{X(s)} = \frac{s+2}{s+5}$$

If the input is a unit step function, find the response of the system.

Solution: If the input is a unit step function, then $X(s) = \frac{1}{s}$ (i.e a pole at the origin). $Y(s) = H(s) X(s)$

$$\text{Therefore } Y(s) = \frac{1}{s} \cdot \frac{s+2}{s+5} = \frac{2/5}{s} + \frac{3/5}{s+5}$$



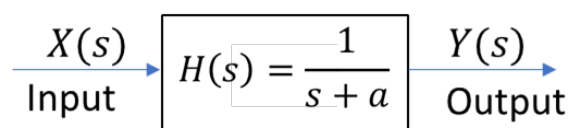
Taking the inverse Laplace transform gives:

$$\therefore y(t) = \frac{2}{5} + \frac{3}{5} e^{-5t} = \frac{2}{5} (1 + \frac{3}{2} e^{-5t})$$

- A pole on the real axis generates an exponential response of the form $e^{-\alpha t}$, where α is the pole location on the real axis.
- The pole at the origin will generate a step function at the output

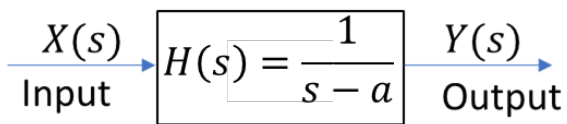
Stability of the System – Real Pole(s)

- A system is **stable** if and only if all its poles lie in the left half of the s -plane.
- **Poles in the left half of the s -plane correspond to exponentially decaying impulse responses (stable systems)**

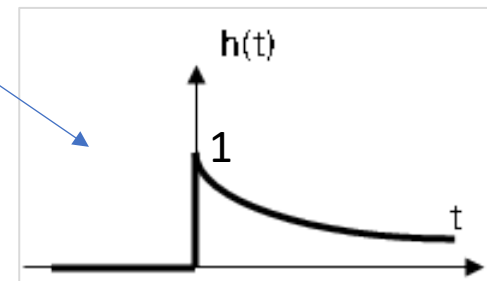
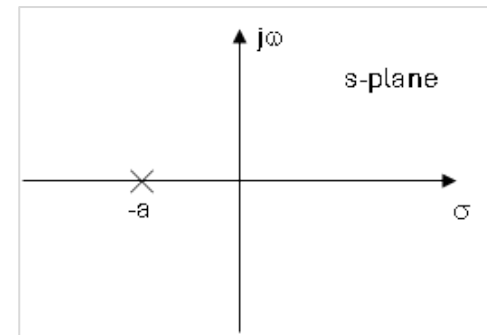
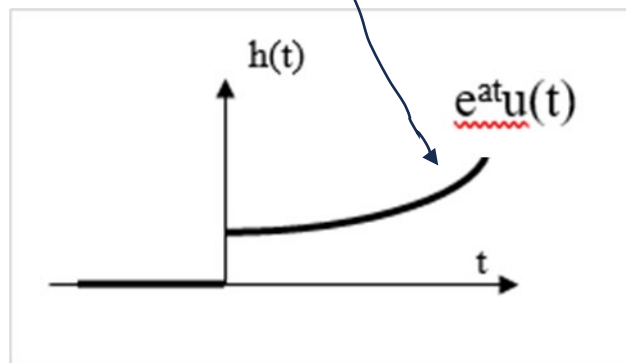
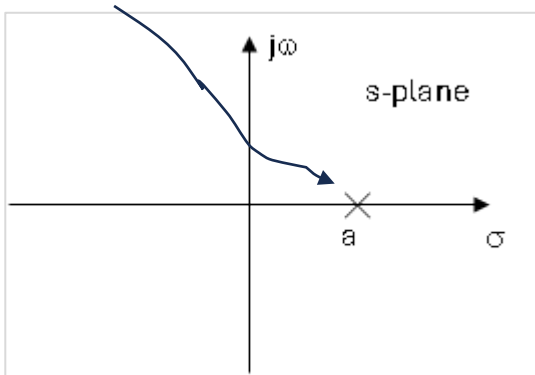


$$H(s) = \frac{1}{s+a} \quad a > 0 \Rightarrow h(t) = e^{-at} u(t)$$

- For a causal rational continuous-time LTI system, Bounded-Input, Bounded-Output (BIBO) stability requires all poles of $H(s)$ to lie strictly in the open left-half plane.
- **Poles in the right half-plane correspond to exponentially growing impulse responses (unstable systems).**



$$H(s) = \frac{1}{s-a} \quad a > 0 \Rightarrow h(t) = e^{at} u(t)$$



$$\Re\{p_i\} < 0 \quad \forall i$$

Stability of the System – Complex Conjugate Poles

- Consider a transfer function with a pair of complex conjugate poles at:

$$H(s) = \frac{1}{(s - p_1)(s - p_2)} = \frac{1}{(s - (\sigma_a + j\omega_a))(s - (\sigma_a - j\omega_a))}$$

$$H(s) = \frac{A}{(s - \sigma_a - j\omega_a)} + \frac{B}{(s - \sigma_a + j\omega_a)}$$

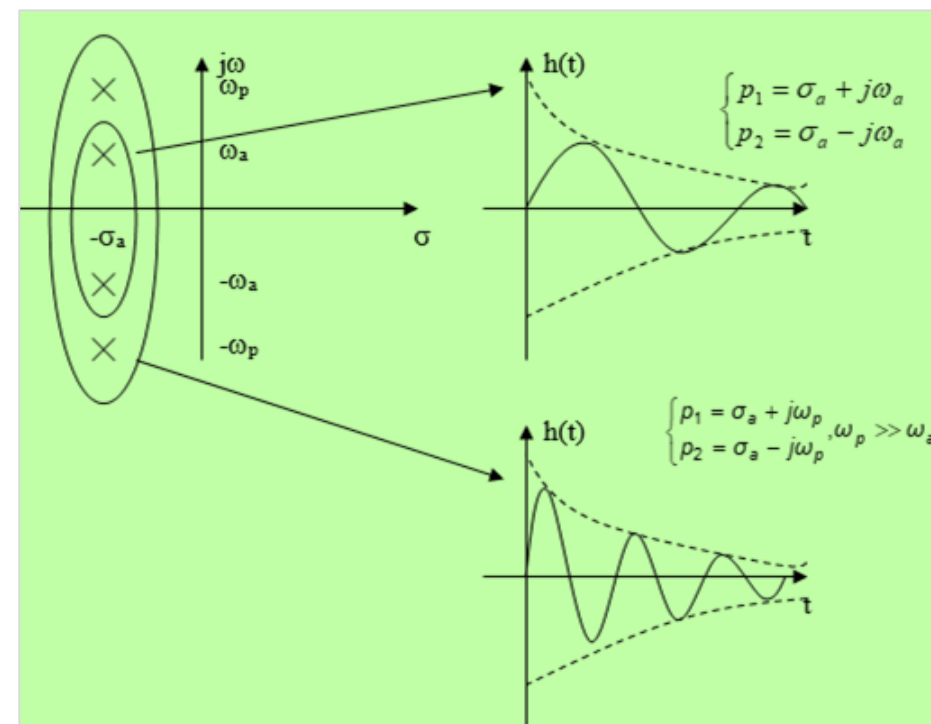
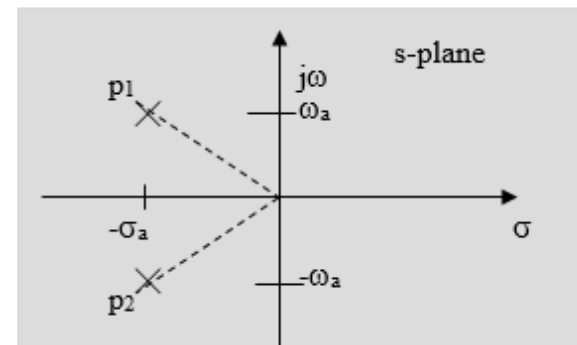
- By evaluating the constants A and B, we obtain

$$H(s) = \frac{\frac{1}{2j\omega_a}}{(s - \sigma_a - j\omega_a)} + \frac{-\frac{1}{2j\omega_a}}{(s - \sigma_a + j\omega_a)}$$

$$h(t) = \mathcal{L}^{-1}[H(s)] = \frac{1}{2j\omega_a} [e^{(\sigma_a + j\omega_a)t} - e^{(\sigma_a - j\omega_a)t}]$$

$$h(t) = \frac{1}{\omega_a} \cdot e^{\sigma_a t} \left[\frac{e^{j\omega_a t} - e^{-j\omega_a t}}{2j} \right] = \frac{1}{\omega_a} e^{\sigma_a t} \sin(\omega_a t)$$

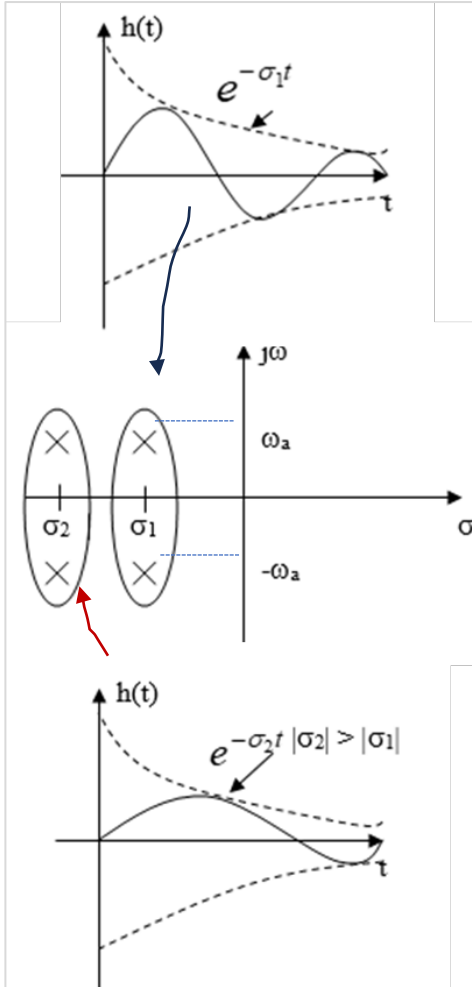
- Impulse response $h(t)$ of a second-order system with complex conjugate poles.
- The real part σ_a determines the decay rate, while the imaginary part ω_a determines the oscillation frequency. Thus, the impulse response depends strongly on the pole locations.



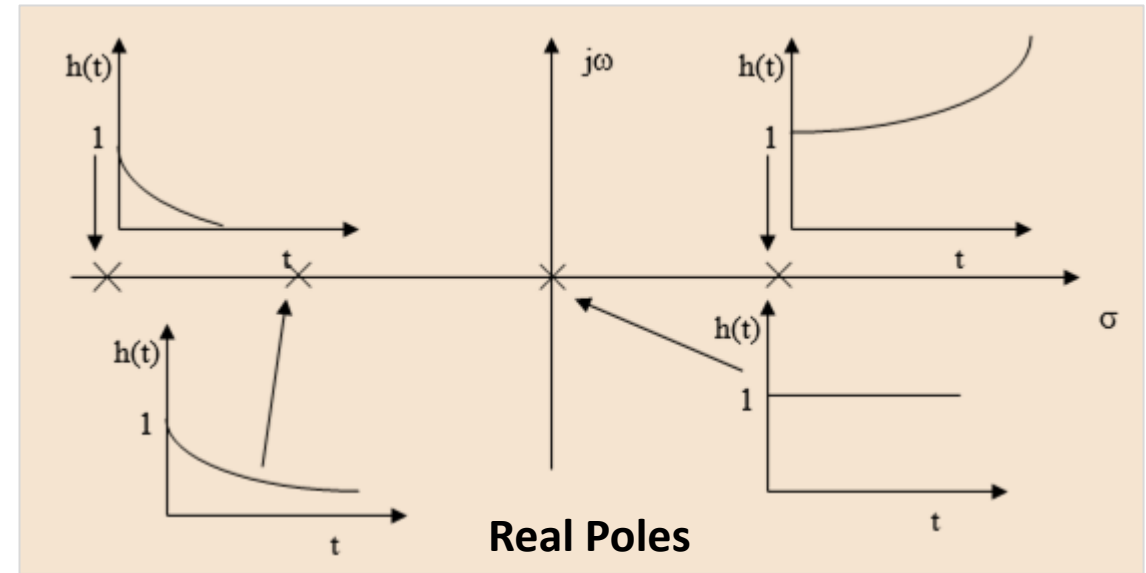
If the real part of the pole is kept constant and the imaginary part increases from ω_a to ω_p , the envelope of the impulse response $h(t)$ remains the same, while the oscillation frequency increases.

Stability of the System

- If the imaginary part is held constant and the real part becomes more negative (moves farther into the left-half plane), the oscillation frequency remains unchanged while the decay rate increases.

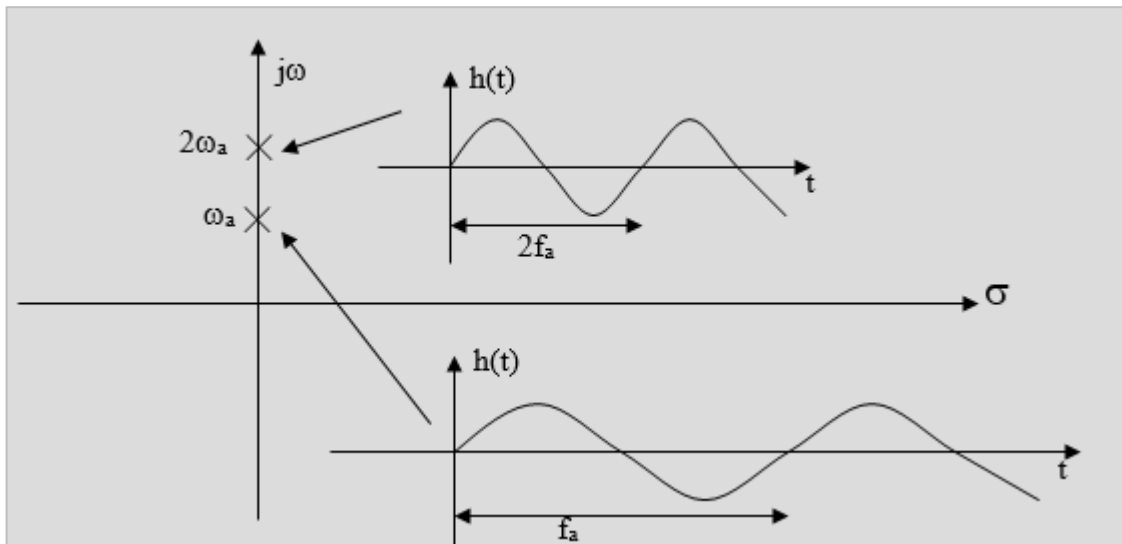


- Notice also that if the pole lies on the real axis (i.e., $\omega_a = 0$) then there is no oscillation in the impulse response.)

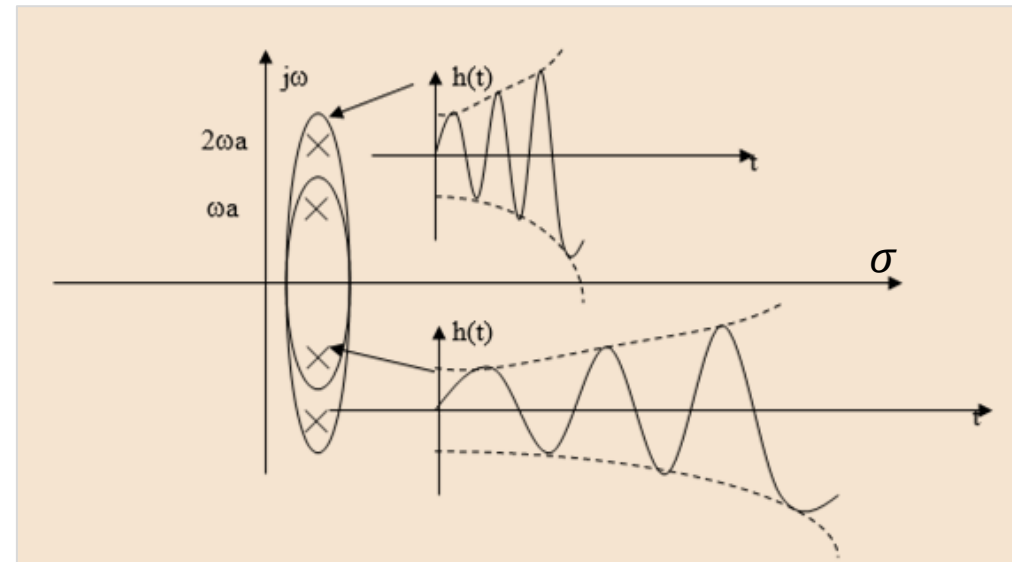


Stability of the System – right hand plane

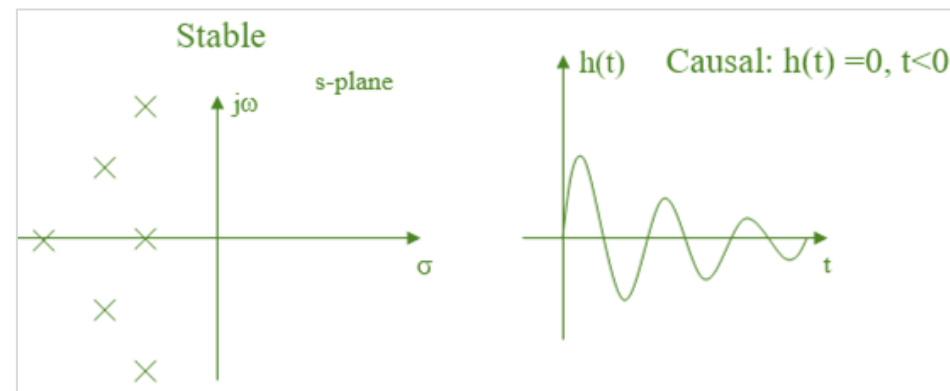
- If the pole lies on the imaginary axis ($\sigma = 0$), there is no decay or growth in the impulse response; it is a pure oscillation.



- If a system has a pole with a positive real part ($\sigma > 0$) its impulse response grows exponentially. Such a system is unstable.

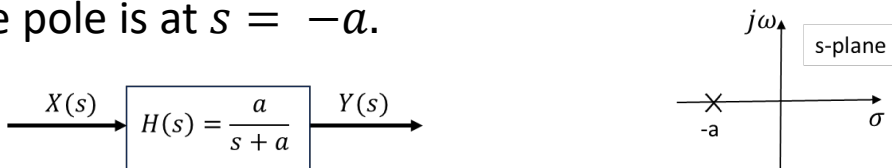


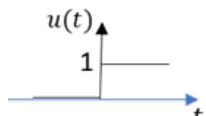
- **Stable and Causal System:** A system that is both causal and stable must have all its poles in the left half-plane of the s-plane.
- A persistent undamped oscillation is not BIBO stable.



Analysis of a first-order system without Zeros

- A first-order system without zeros can be described by the transfer function: $H(s) = \frac{a}{s+a}$, the pole is at $s = -a$.



- If the input is a unit step function () where $X(s) = \frac{1}{s}$, the Laplace transform of the step response is:

$$Y(s) = X(s) \cdot H(s) = \frac{1}{s} \frac{a}{s+a} = \frac{1}{s} - \frac{1}{s+a}.$$

Taking the inverse transform of $Y(s)$, the step response is given by:

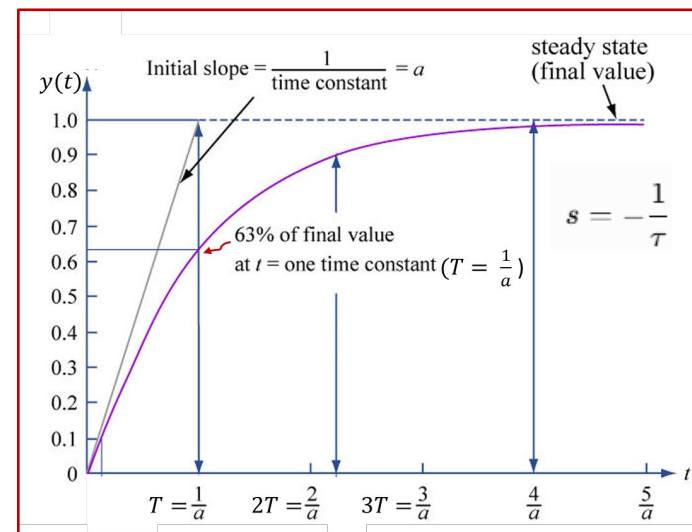
$$y(t) = 1 - e^{-at} \quad (\text{see figure on the right})$$

- We will now examine the significance of parameter a , the only parameter needed to describe the response.

$$\text{When } t = 1/a, y(t)|_{t=1/a} = 1 - e^{-a \cdot (1/a)} = 0.63$$

- We call $1/a$ the time constant (T) of the response. That is the time constant (T) is the time it takes for the step response to rise to 63% of its final value.
- The reciprocal of the time constant has the units (1/seconds) or frequency. Thus, we call the parameter a the *cutoff frequency*.

- The time constant $T = 1/a$ is related to (transient response) the speed at which the system responds to a step input.
- Since the pole of the transfer function is at $s = -a$, we can say the pole is located at the negative of the reciprocal of the time constant and further the pole from the imaginary axis the faster the transient response.



Analysis of a second-order system without zeros

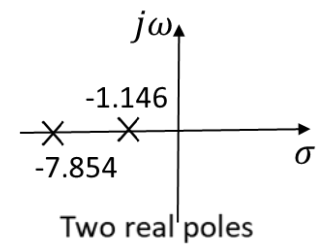
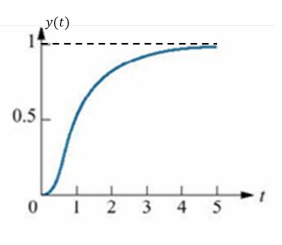
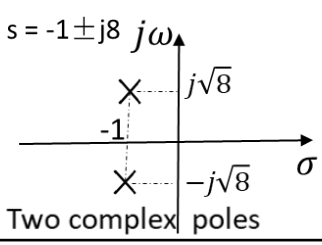
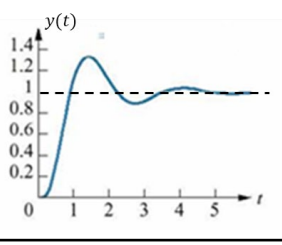
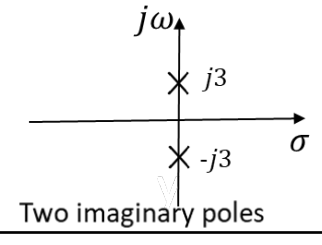
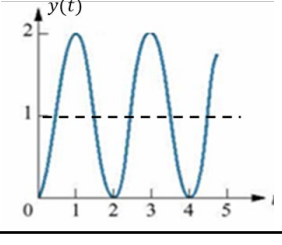
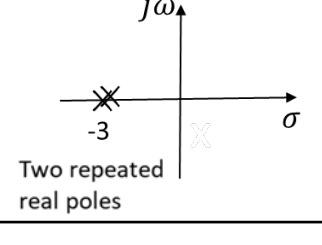
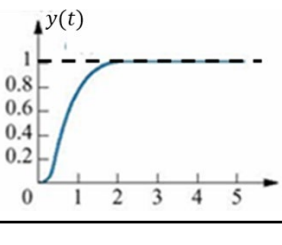
- Compared to a first order system, a second-order system exhibits a wide range of responses.
- In order to become familiar with a second order system responses, we will look at some examples.
- Let's look at a general case of a second order transfer function:

$$H(s) = \frac{Y(s)}{X(s)} = \frac{a}{s^2 + bs + a} \quad \begin{array}{c} X(s) \rightarrow \boxed{H(s)} \rightarrow Y(s) \end{array}$$

- By assigning appropriate values of a and b , we can show all possible second order transient responses. The term a in the numerator is simply a multiplying factor that can take any values.
- If the input $X(s)$ is a unit step input ($X(s) = \frac{1}{s}$), then the output $Y(s) = X(s) \cdot H(s) = \frac{1}{s} \frac{a}{s^2 + bs + a}$
- $\therefore y(t) = \mathcal{L}^{-1} \left[\frac{1}{s} \frac{a}{s^2 + bs + a} \right]$
- We will consider four cases where a and b will take different values.

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Overdamped $\zeta > 1$, Critically damped $\zeta = 1$, Underdamped $0 < \zeta < 1$, Undamped $\zeta = 0$.

	Transfer Function, $H(s)$	s-plane	Unit step response, $y(t)$
Case 1: $a = 9$ $b = 9$	Overdamped $H(s) = \frac{9}{s^2 + 9s + 9}$ $y(t) = 1 + 0.171e^{-7.854t} - 1.171e^{-1.146t}$	 Two real poles	
Case 2: $a = 9$ $b = 2$	Underdamped $H(s) = \frac{9}{s^2 + 2s + 9}$ $y(t) = 1 - e^{-t}(\cos\sqrt{8}t + \frac{\sqrt{8}}{8}\sin\sqrt{8}t)$	 Two complex poles	
Case 3: $a = 9$ $b = 0$	Undamped $H(s) = \frac{9}{s^2 + 9}$ $y(t) = 1 - \cos(3t)$	 Two imaginary poles	
Case 4: $a = 9$ $b = 6$	Critically damped $H(s) = \frac{9}{s^2 + 6s + 9}$ $y(t) = 1 - 3te^{-3t} - e^{-3t}$	 Two repeated real poles	

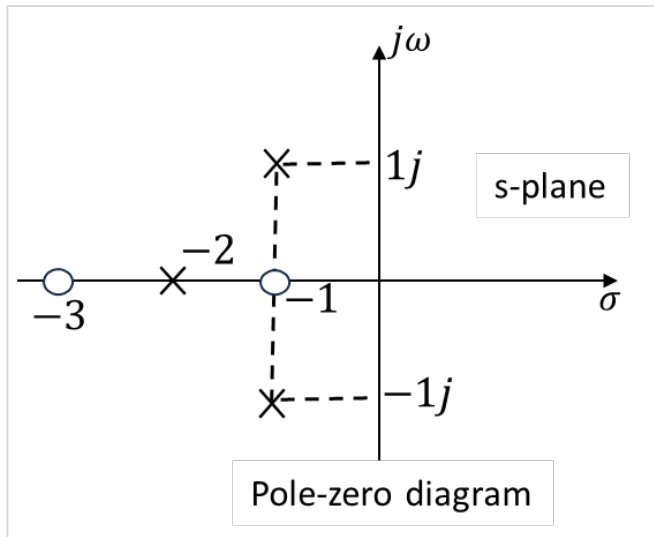
Pole-zero Plots

Example: Draw the pole zero plot of the transfer function $H(s) = \frac{(s+1)(s+3)}{[(s+1)^2+1](s+2)}$

Solution:

Zeros: $s = -1$ and $s = -3$

Poles: $s = -2$ and $s^2 + 2s + 2 = 0 \rightarrow s = -1 \pm j$



1. Factor numerator
2. Factor denominator
3. Find zeros
4. Find poles
5. Plot on s-plane
6. Interpret behaviour

Example: A transfer function $H(s)$ has **poles** at $s = 0, -2$ and -4 , with **zeros** at $s = -1$ and $s = -3$. If $H(1) = \frac{4}{3}$, determine $H(s)$. Draw the pole-zero plot of the transfer function.

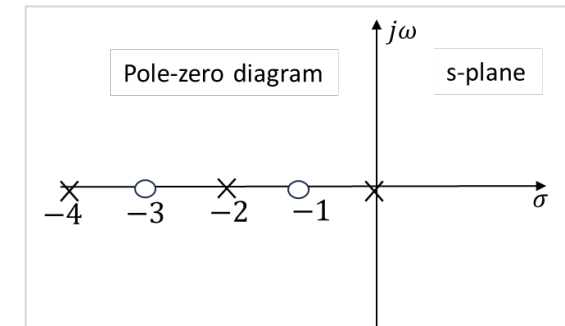
Solution:

$$H(s) = K \frac{(s+1)(s+3)}{s(s+2)(s+4)}$$

Since $H(1) = \frac{4}{3}$, evaluating $H(s)$ at $s = 1$ yields

$$\frac{4}{3} = H(1) = K \frac{(1+1)(1+3)}{1(1+2)(1+4)} \rightarrow k = 2.5$$

$$\therefore H(s) = 2.5 \frac{(s+1)(s+3)}{s(s+2)(s+4)}$$



Cascade and Parallel Structures

Block Diagrams

- A linear system can be characterised by its transfer function $H(s)$, where $X(s)$ and its output $Y(s)$ are shown in Figure 1(a).
- When the transfer function of overall system $H(s)$ is the product of two transfer functions, then it is called cascade form. (see figure 1(b)).

$$H(s) = H_1(s) \cdot H_2(s)$$

- When two transfer functions $H_3(s)$ and $H_4(s)$, appear in parallel, (see Figure 1c), then the overall transfer function is given by

$$H(s) = H_3(s) + H_4(s)$$

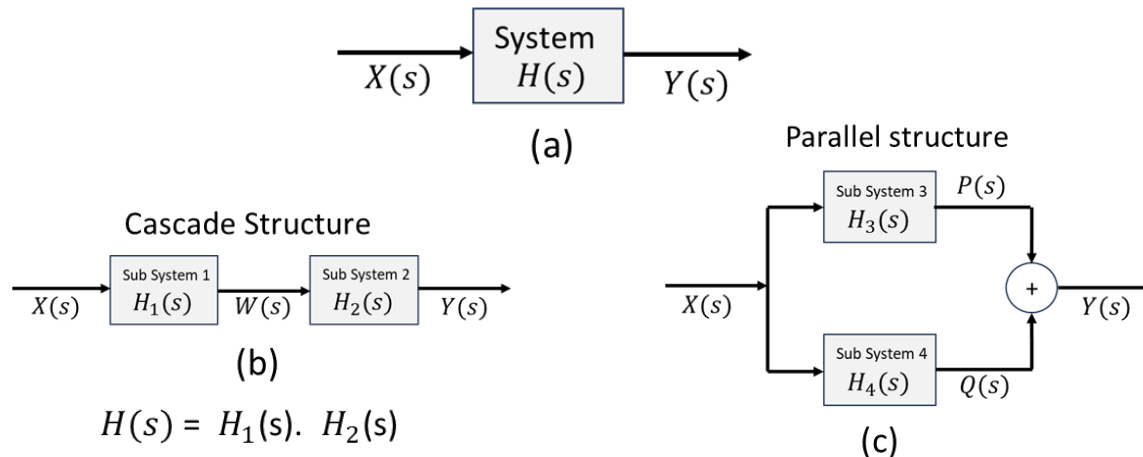


Figure 1

$$H(s) = H_3(s) + H_4(s)$$

Example

- Consider the transfer function $H(s) = \frac{4s+28}{s^2+6s+5}$ and this can be expressed in cascade and parallel form.

Cascade form:

$$H(s) = \frac{4s+28}{s^2+6s+5} = \underbrace{\left(\frac{4s+28}{s+1}\right)}_{H_1(s)} \underbrace{\left(\frac{1}{s+5}\right)}_{H_2(s)} \quad \{\text{Perform Factorisations}\}$$

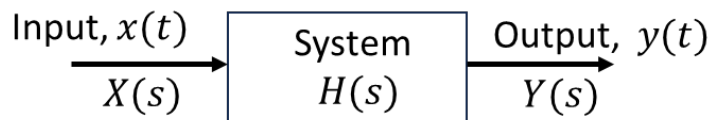
Parallel form:

$$H(s) = \frac{4s+28}{s^2+6s+5} = \underbrace{\left(\frac{6}{s+1}\right)}_{H_3(s)} - \underbrace{\left(\frac{2}{s+5}\right)}_{H_4(s)} \quad \{\text{Perform Partial fractions}\}$$

The Frequency Response

Magnitude and Phase Responses

- If the input of a linear system is $x(t)$ and the output is $y(t)$ then as explained before the Laplace transform of the output $Y(s) = H(s) \cdot X(s)$ where $H(s)$ the transfer function and $X(s)$ the Laplace transform of the input $x(t)$.



- The steady-state response ($\sigma = 0$) of a linear system to a sinusoidal input say $x(t) = \sin(\omega t)$, is obtained from the expression for $H(s)$ by replacing s with $j\omega$. The function $H(j\omega) \{= H(\omega)\}$ (is called the **frequency response** function).
- The **modulus** of $H(j\omega)$ gives the magnitude of the sinusoidal response, $|H(j\omega)| \{= |H(\omega)|\}$, (relative to the magnitude of the input sinusoid).
- The **angle** of $H(j\omega)$ gives the phase of the response, $\phi(\omega)$ (relative to the input sinusoid).

Magnitude and Phase Responses

Example: The magnitude response is a plot of the variation of the modulus of $H(j\omega)$ with frequency. Consider the transfer function:

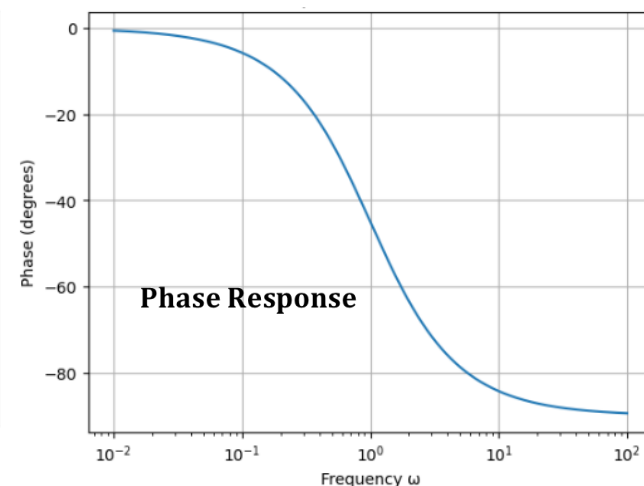
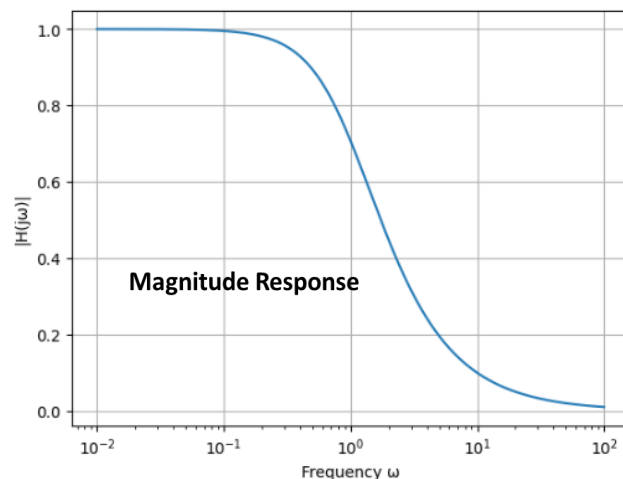
$$H(s) = \frac{a}{s+a}$$

Solution:

$$H(j\omega) = H(s)|_{s=j\omega} = \frac{a}{j\omega+a}; \quad \text{Magnitude: } |H(j\omega)| = \frac{a}{\sqrt{\omega^2+a^2}};$$

$$\text{Phase angle: } \phi(\omega) = \arg \left\{ \frac{a}{j\omega+a} \right\} = \angle H(j\omega) = -\tan^{-1}\left(\frac{\omega}{a}\right)$$

- At $\omega = 0$: $|H(j\omega)| = 1$, phase = 0°
- At $\omega = a$: $|H(j\omega)| = \frac{1}{\sqrt{2}}$, phase = -45°
- As $\omega \rightarrow \infty$: $|H(j\omega)| \rightarrow 0$, phase $\rightarrow -90^\circ$



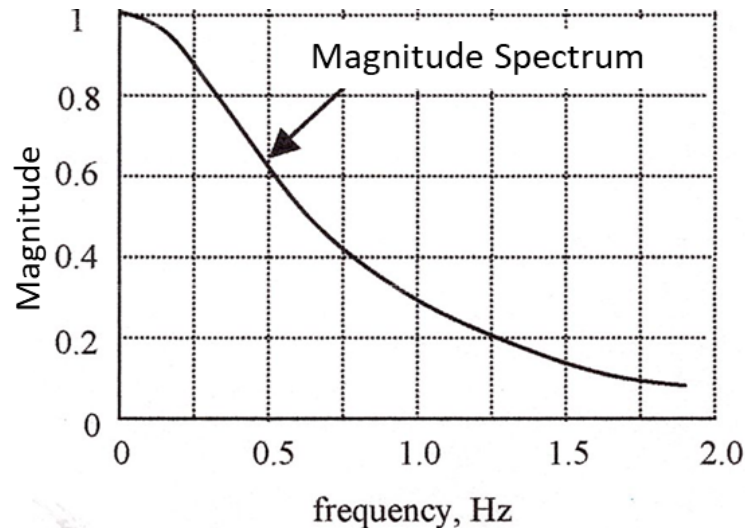
The Frequency Response

Magnitude Response

Example: The magnitude response is a plot of the variation of the modulus of $G(j\omega)$ with frequency. Consider the transfer function:

$$H(s) = \frac{16}{(s+4)^2} \rightarrow H(j\omega) = \frac{16}{(j\omega+4)^2} \therefore |H(j\omega)| = \frac{16}{\omega^2+16}$$

- The magnitude spectrum (Magnitude vs frequency), below shows the output is approximately equal to the input at very low frequencies less than 0.1 Hz, but the magnitude decreases as the frequency increases.



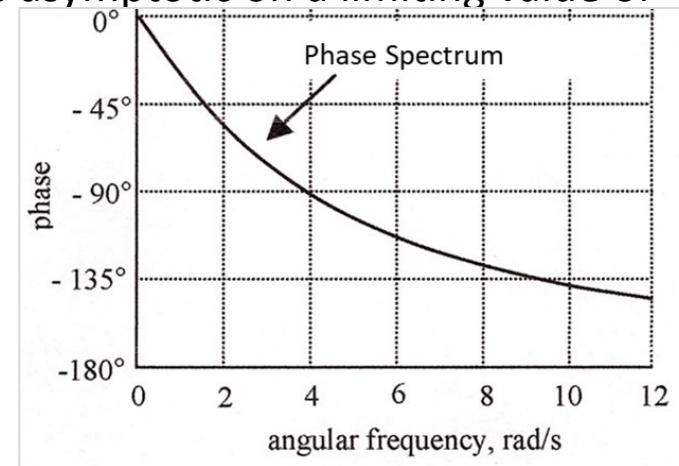
Low pass
filter

Phase Response

The phase response of the system is:

$$\phi(\omega) = \arg \left\{ \frac{16}{(j\omega+4)^2} \right\} = -2 \tan^{-1} \left(\frac{\omega}{4} \right)$$

- The phase response shown below shows that the output will be almost in phase with the input sinusoid only at very low frequencies.
- As the frequency increases, the phase lag increases
- Note that phase lag does not increase indefinitely. It is asymptotic on a limiting value of -180° .



Bandwidth (Lowpass Filter)

Bandwidth: The frequency at which the magnitude spectrum falls to $\frac{1}{\sqrt{2}}$ (≈ 0.707) of its maximum value (or its peak value).

For a lowpass filter or system: $H(s) = \frac{a}{s+a}$

the magnitude is

$$|H(j\omega)| = \frac{a}{\sqrt{a^2 + \omega^2}}$$

The cutoff occurs when

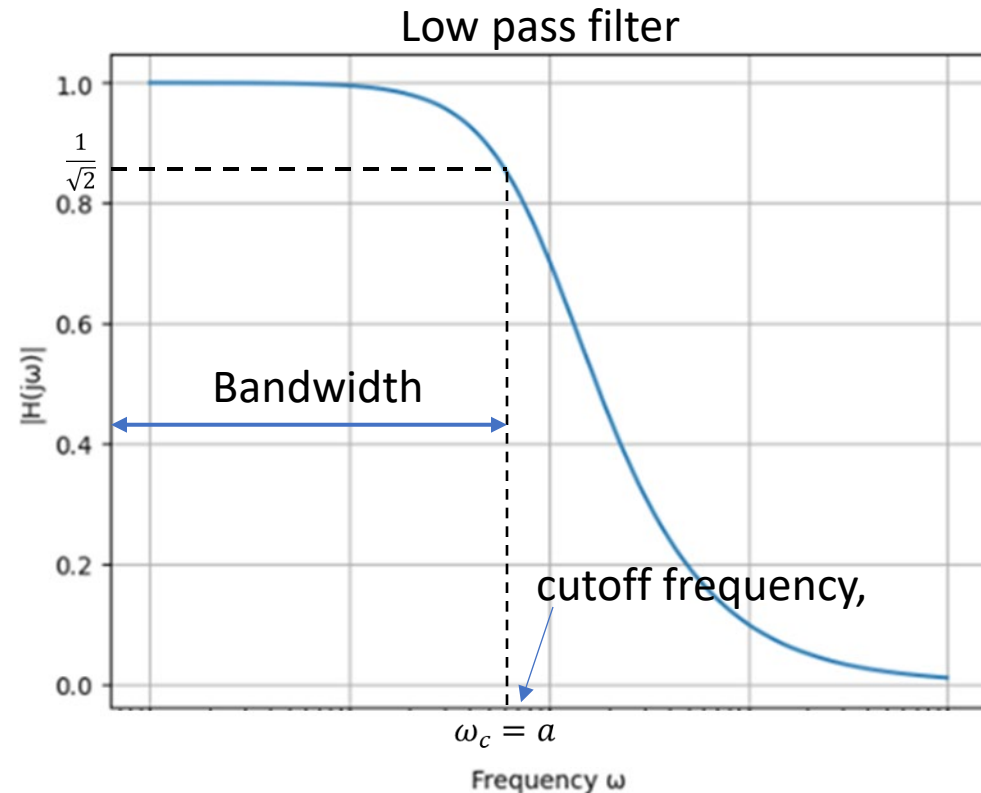
$$|H(j\omega)| = \frac{1}{\sqrt{2}}$$

which gives

$$\omega = a = \omega_c \text{ (known as cut-off frequency)}$$

So the bandwidth of this system is

$$BW = a$$

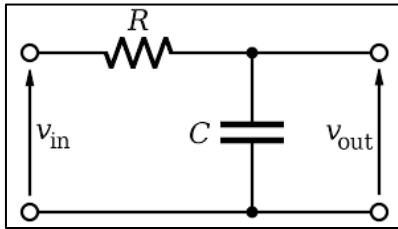


The low-pass filter passes all frequencies below a certain value, ω_c , called the cutoff frequency,

Bandwidth (Lowpass Filter)

Example: Design a first-order low-pass filter with a cutoff frequency of 1 kHz, using a resistor ($R = 1.6 \text{ k}\Omega$) and a capacitor (C).

1. Write the transfer function $H(s)$ of the RC low pass filter.
2. Calculate the required capacitor value C .
3. Sketch the magnitude response of $H(s)$

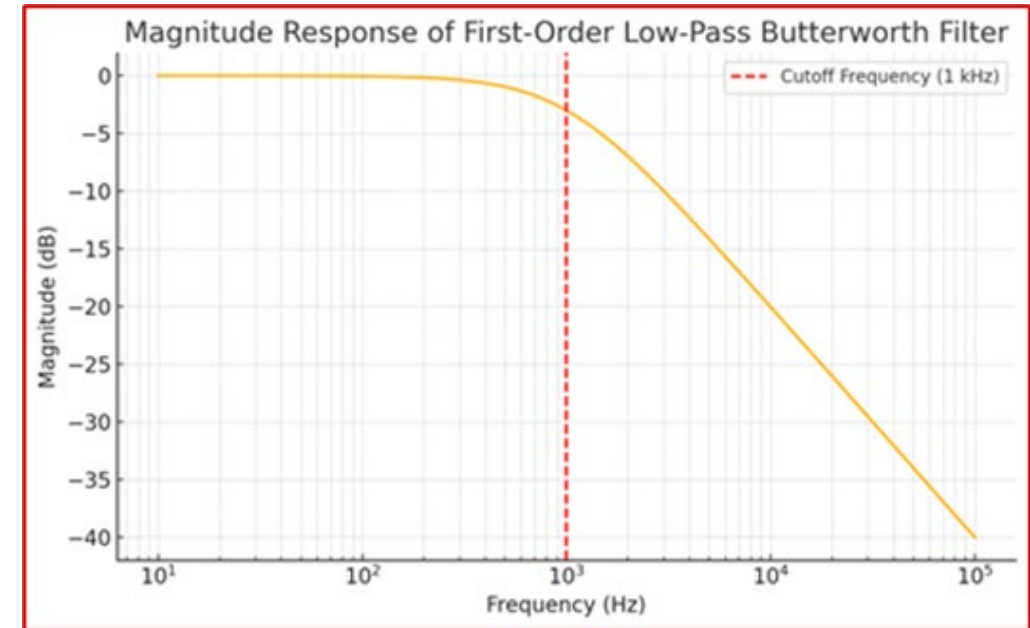


Model Answer:

The transfer function $H(s)$ is given by: $H(s) = \frac{v_{out}(s)}{v_{in}(s)} = \frac{\frac{1}{Cs}}{R + \frac{1}{Cs}} = \frac{1}{1 + RCs}$

$$H(s) = \frac{1/RC}{s + 1/RC}; \quad H(\omega) = \frac{1/RC}{j\omega + 1/RC}; \quad |H(j\omega)| = \frac{1/RC}{\sqrt{\omega^2 + (1/RC)^2}}$$

The cut off frequency $f_c = \frac{1}{2\pi RC}$ and $c = \frac{1}{2\pi \cdot 1600 \cdot 1000} = 99.47 \text{ nF}$



Decibels

Decibels

Definition: The decibel (dB) is a logarithmic measurement unit for sound levels and finds extensive application in electronics, signal processing, and communication.

- It represents ratios such as power, sound pressure, voltage, intensity, and more using a logarithmic scale:

$$\text{decibels}(dB) = 20 \log_{10} \frac{V_{out}}{V_{in}} = 10 \log_{10} \left(\frac{V_{out}}{V_{in}} \right)^2 = 10 \log_{10} \frac{P_{out}}{P_{in}}$$

Common values:

- $10 \log_{10} 2 = 3 \text{ dB}$;
- $10 \log_{10} \frac{1}{2} = -3 \text{ dB}$ (half power)
- $10 \log_{10} 10 = 10 \text{ dB}$; $10 \log_{10} 100 = 20 \text{ dB}$; $10 \log_{10} 1000 = 30 \text{ dB}$]
(10 dB for every factor of 10 in power)

Sounds	Level
Faint	20dB (A faint Whisper is 30dB)
Soft (Quiet)	30dB
Moderate	40dB (normal conversation)
Loud	80dB (alarm clocks, vacuum cleaners)
Very Loud	90dB(Blenders);110dB (Concerts, car horns)
Uncomfortable	120dB (jet planes during take off)
Painful and dangerous	130dB(Jackhammers); 140dB(Gunshots) *Use hearing protection

- ✓ Over 85 dB for extended periods can cause permanent hearing loss
- ✓ Zero decibels (0 dB) represent the absolute threshold of human hearing, below which we cannot hear a sound.

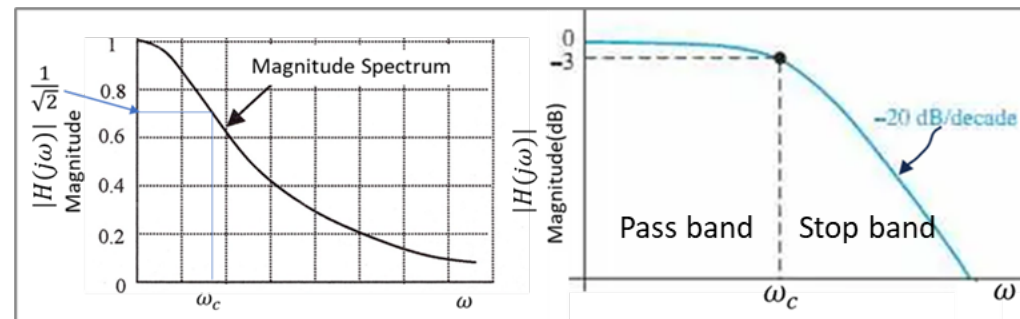
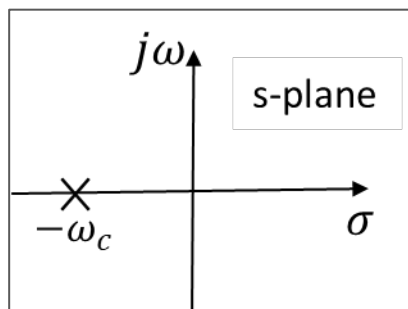
1st order low-pass filter

1st order low-pass filter

$$H(s) = \frac{\omega_c}{s + \omega_c}; \quad H(j\omega) = \frac{\omega_c}{j\omega + \omega_c}$$

$|H(j\omega)|$ - Magnitude

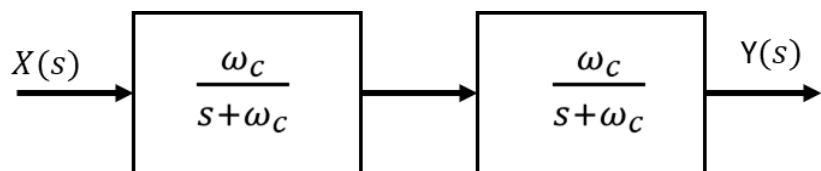
ω_c - 3dB cutoff frequency;



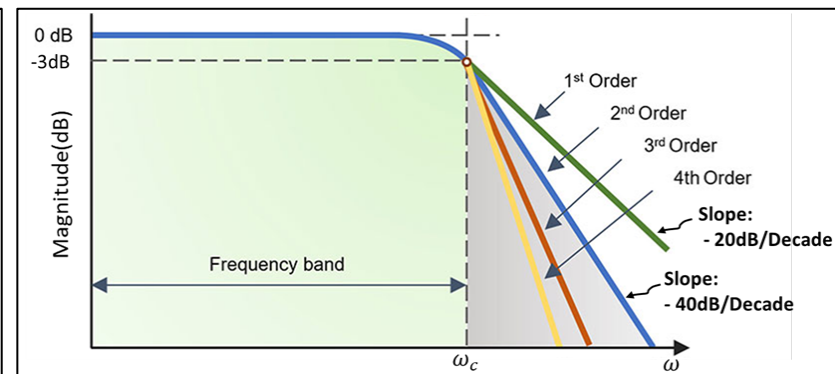
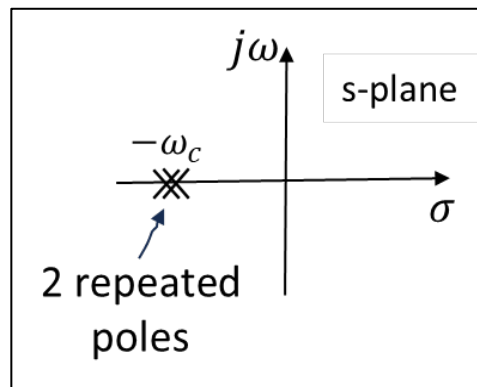
The **cutoff frequency** is typically defined as the frequency where the magnitude drops by **3 dB (or $1/\sqrt{2}$)** from the passband level.

2nd order low-pass filter

Note: A second-order filter can be constructed by cascading two first-order filters



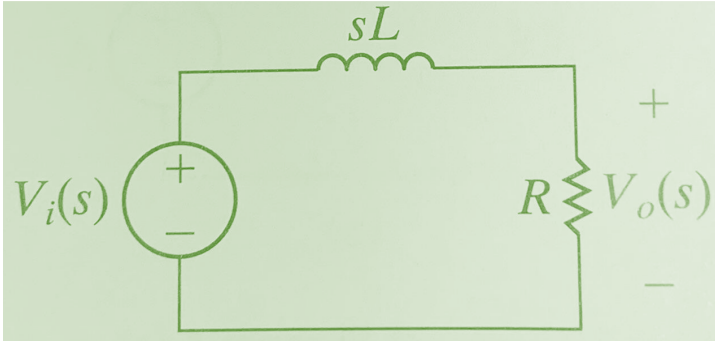
Cascading two first-order low-pass stages produces a second-order low-pass system with two real poles.



- ✓ Both first-order and second-order filters can have the same cutoff frequency.
- ✓ Each pole contributes approximately -20 dB/decade to the high-frequency asymptotic slope.
- ✓ A first-order filter rolls off at 20 dB/decade beyond the cutoff and a second-order filter rolls off at 40 dB/decade asymptotically for low-pass systems with two more poles than zeros, meaning it attenuates unwanted frequencies more sharply.

1st order low-pass filter

Series RL circuit



The voltage transfer function for this circuit is

$$H(s) = \frac{R/L}{s + R/L}.$$

To find the frequency response, we substitute $s = j\omega$

$$H(j\omega) = H(\omega) = \frac{R/L}{j\omega + R/L}$$

Magnitude response: $|H(\omega)| = \frac{R/L}{\sqrt{\omega^2 + (R/L)^2}},$

Phase Response:

$$\phi(\omega) = \text{Arg}(H(\omega)) = -\tan^{-1}\left(\frac{\omega L}{R}\right)$$

When $\omega = 0$ (DC component), the denominator and the numerator are equal and $|H(0)| = 1$.

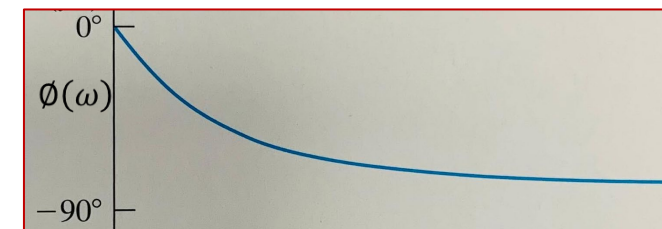
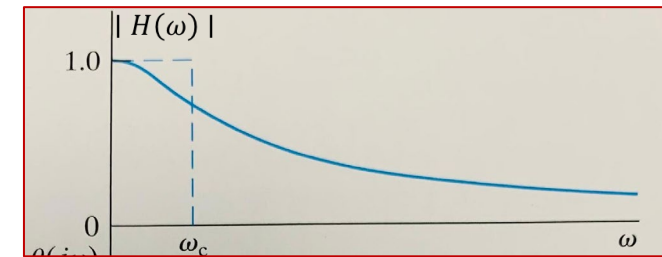
The cutoff frequency, ω_c is defined as the frequency at which

$$|H(\omega = \omega_c)| = \frac{1}{\sqrt{2}} H_{max}.$$

For the low-pass filter, $H_{max} = |H(j0)| = 1$

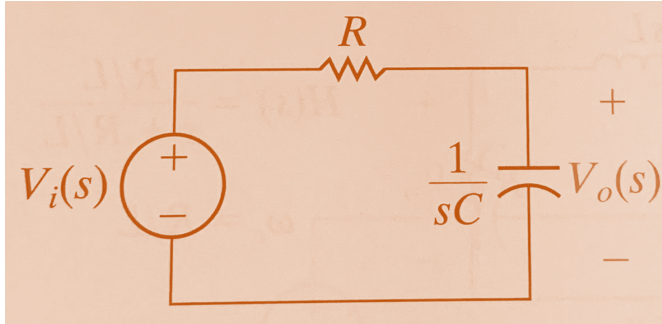
$$|H(\omega_c)| = \frac{1}{\sqrt{2}} |1| = \frac{R/L}{\sqrt{\omega_c^2 + (R/L)^2}}$$

Solving the above equation for ω_c , we get $\omega_c = \frac{R}{L}$



1st order low-pass filter

Series RC circuit



The voltage transfer function for this circuit is

$$H(s) = \frac{1}{s + \frac{1}{RC}}.$$

To find the frequency response, we substitute $s = j\omega$

$$H(j\omega) = H(\omega) = \frac{\frac{1}{RC}}{j\omega + \frac{1}{RC}}$$

Magnitude response: $|H(\omega)| = \frac{\frac{1}{RC}}{\sqrt{\omega^2 + \left(\frac{1}{RC}\right)^2}}$

Phase Response:

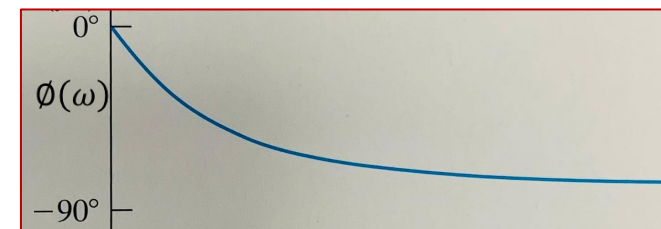
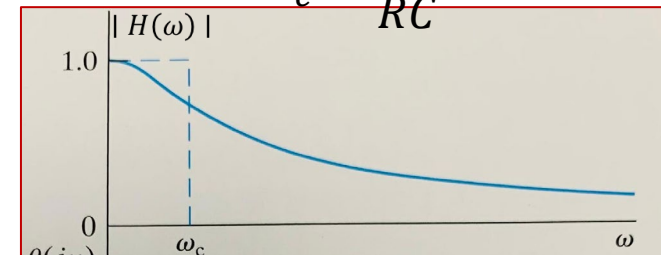
$$\phi(\omega) = \text{Arg}(H(\omega)) = -\tan^{-1}(\omega RC)$$

At the cutoff frequency ω_c , $|H(\omega)|$ is equal to $(1/\sqrt{2})H_{max}$.
For a low-pass filter,

$$H_{max} = H(0) = 1.$$
$$\therefore |H(j\omega_c)| = \frac{1}{\sqrt{2}}(1) = \frac{\frac{1}{RC}}{\sqrt{\omega_c^2 + \left(\frac{1}{RC}\right)^2}}.$$

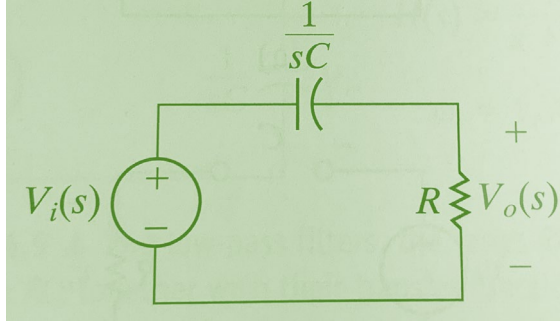
Solving this equation for ω_c , we get

$$\omega_c = \frac{1}{RC}$$



1st order High-pass filter

Series RC circuit



The voltage transfer function for this circuit is

$$H(s) = \frac{s}{s + \frac{1}{RC}}$$

To find the frequency response, we substitute $s = j\omega$

$$H(j\omega) = H(\omega) = \frac{j\omega}{j\omega + \frac{1}{RC}}$$

Magnitude response: $|H(\omega)| = \frac{\omega}{\sqrt{\omega^2 + \left(\frac{1}{RC}\right)^2}}$

Phase Response:

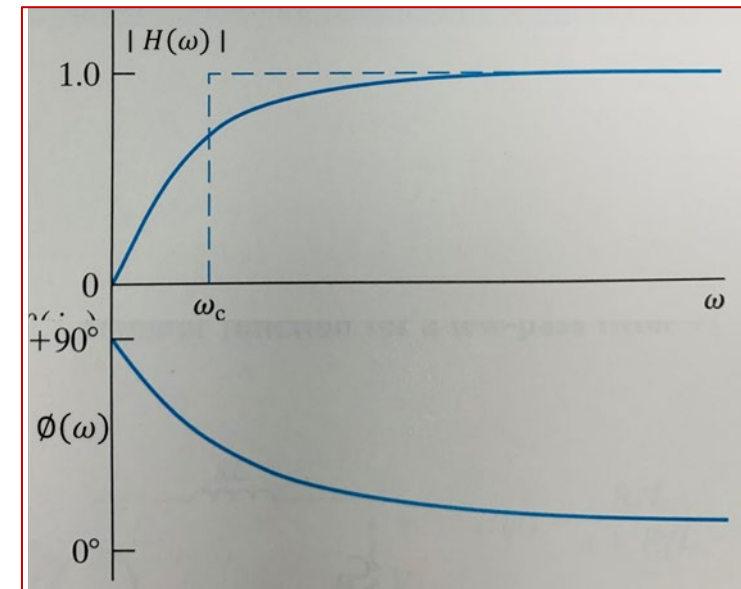
$$\phi(\omega) = \text{Arg}(H(\omega)) = 90^\circ - \tan^{-1}(\omega RC)$$

At the cutoff frequency, the magnitude of the transfer function is $(1/\sqrt{2})H_{max}$

For a high-pass filter,

$$H_{max} = |H(j\omega)|_{\omega=\infty} = |H(j\infty)| = 1$$
$$\frac{1}{\sqrt{2}} = \frac{\omega_c}{\sqrt{\omega_c^2 + \left(1/RC\right)^2}}$$

Solving this equation for ω_c , we get $\omega_c = \frac{1}{RC}$

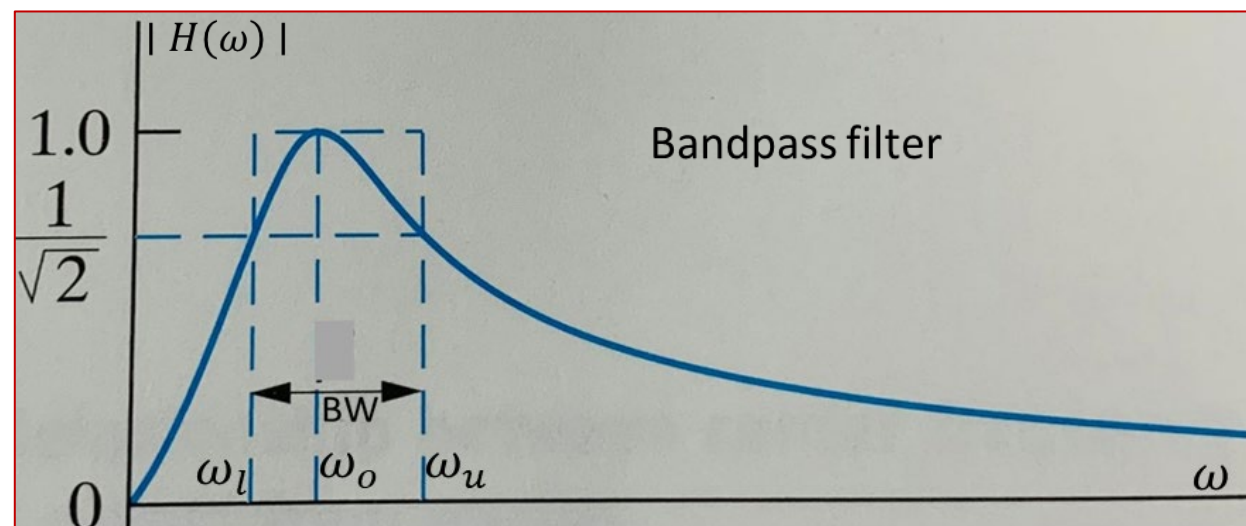


Bandpass filters

Centre Frequency, Bandwidth and quality Factor

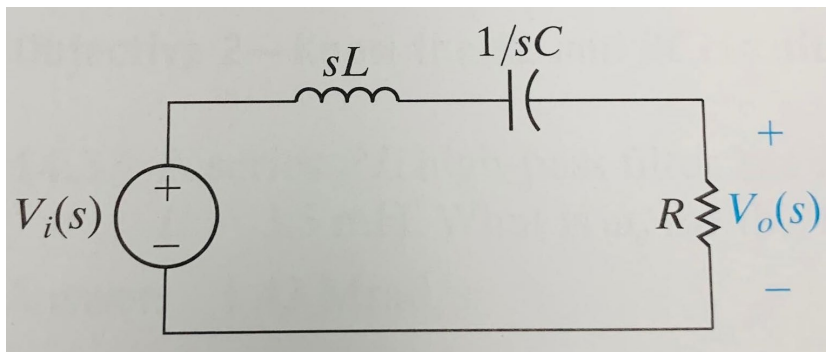
- A Bandpass filter has two cutoff frequencies , ω_u and ω_l which identify the passband.
- The cut off frequencies are defined as the frequencies for which the magnitude of the transfer function equals $(1/\sqrt{2})H_{max}$.
- There are three important parameters that characterize a bandpass filter. The first is the **centre frequency**, ω_o defined as the frequency for which a circuit's transfer function is purely real.
- Another name for the centre frequency is the **resonant frequency**.
- The centre frequency is the geometric centre of the passband, that is,
$$\omega_o = \sqrt{\omega_u \omega_l}$$
- For bandpass filters , the magnitude of the transfer function is maximum at the Centre frequency (Resonant frequency) $\{(H_{max} = |H(\omega_o)|)\}$

- The second parameter is the **bandwidth (BW)**, which is the width of the passband. $(BW = \omega_u - \omega_l)$
- The final parameter is the **quality factor (Q)**, which is the ratio of the centre frequency to the bandwidth. $Q = \frac{\omega_o}{BW}$
- The quality factor gives a measure of the width of the passband, independent of its location on the frequency axis. It also describes the shape of the magnitude plot, independent of frequency.



Bandpass filters

Series RLC circuit



The voltage transfer function for this circuit is

$$H(s) = \frac{(R/L)s}{s^2 + (R/L)s + (1/LC)}$$

To find the frequency response, we substitute $s = j\omega$

$$H(\omega) = (H(j\omega)) = \frac{\left(\frac{R}{L}\right)j\omega}{(j\omega)^2 + \left(\frac{R}{L}\right)j\omega + \left(\frac{1}{LC}\right)}$$

Magnitude Response:

$$|H(j\omega)| = \frac{\omega(R/L)}{\sqrt{[(1/LC) - \omega^2]^2 + [\omega(R/L)]^2}}$$

$$\phi(\omega) = \text{Arg}(H(\omega)) = 90^\circ - \tan^{-1} \left[\frac{\omega(R/L)}{(1/LC) - \omega^2} \right].$$

- the centre frequency, ω_o , is defined as the frequency for which the circuit's transfer function is purely real.
- This is true for the standard second-order RLC bandpass considered here, resonance occurs when the reactive terms cancel, and the transfer function becomes purely real.
- The transfer function for the RLC circuit will be real when the frequency of the voltage source makes the sum of the capacitor and inductor impedances zero:

$$j\omega_o L + \frac{1}{j\omega_o C} = 0.$$

- Solving the above for ω_o , we get

$$\omega_o = \sqrt{\frac{1}{LC}}$$

- The five parameters that characterise the RLC band-pass filter can be easily calculated.

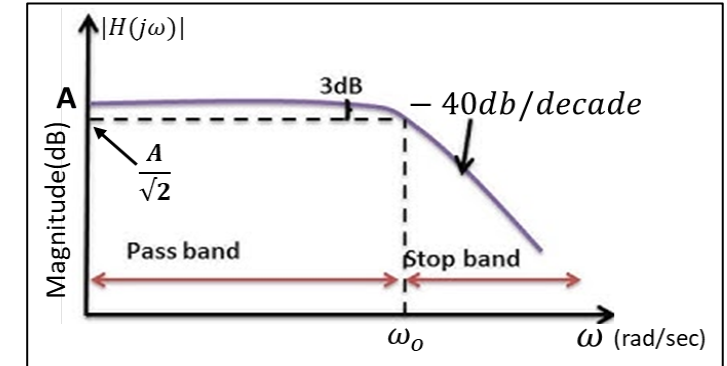
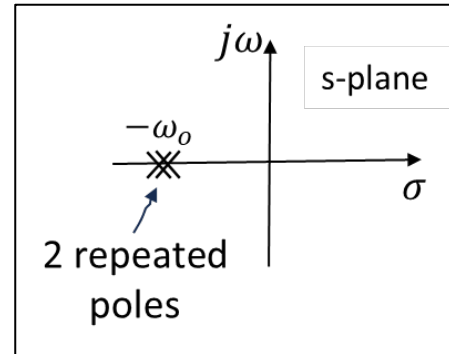
2nd order lowpass filter

2nd order lowpass filter (real repeated poles)

$$H(s) = \frac{\omega_o^2}{s^2 + b_o s + \omega_o^2} \leftarrow \text{Real poles *repeated*}$$

$$b_o \text{ (bandwidth)} = \frac{\omega_o}{Q}; \quad \omega_o - \text{cutoff frequency}$$

Q is the quality factor

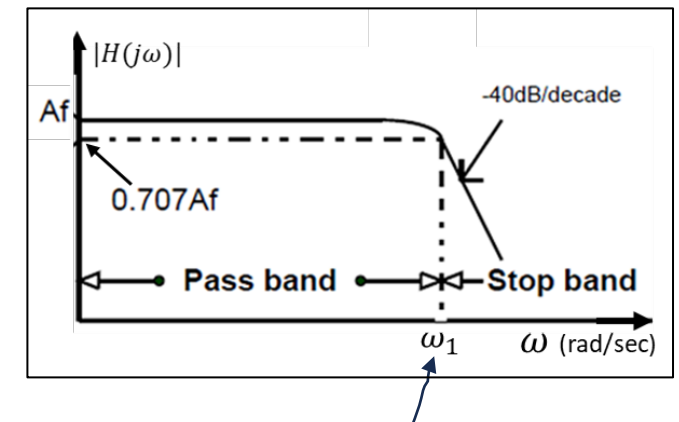
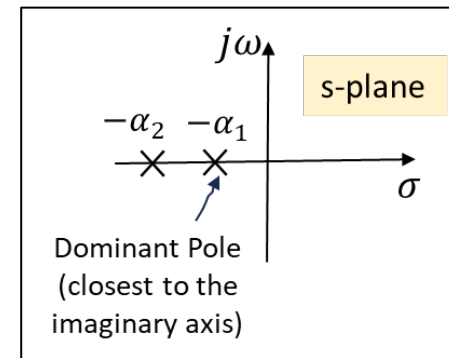


2nd order lowpass filter (real distinct poles)

$$H(s) = \frac{\omega_o^2}{s^2 + b_o s + \omega_o^2} \leftarrow \text{Real *distinct* poles}$$

$$b_o \text{ (bandwidth)} = \frac{\omega_o}{Q};$$

Q is the quality factor



Cut-off frequency (ω_1) is determined by the dominant pole α_1

2nd order resonant low-pass filter

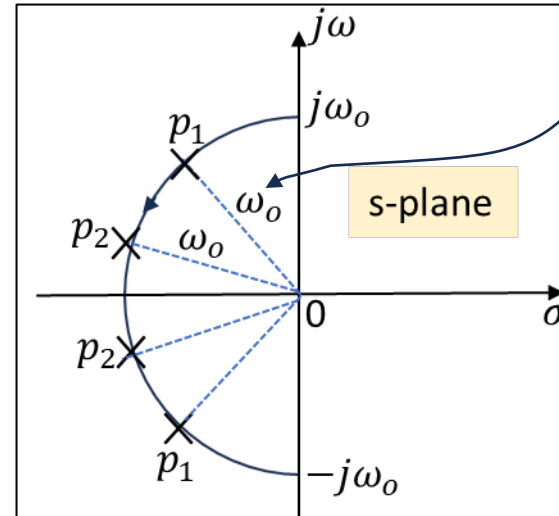
2nd order resonant low-pass filter

$$H(s) = \frac{\omega_o^2}{s^2 + b_o s + \omega_o^2}$$

← *Complex Conjugate Poles*

$$b_o(\text{bandwidth}) = \frac{\omega_o}{Q}; \text{ Q is the quality factor}$$

ω_o is the resonant frequency

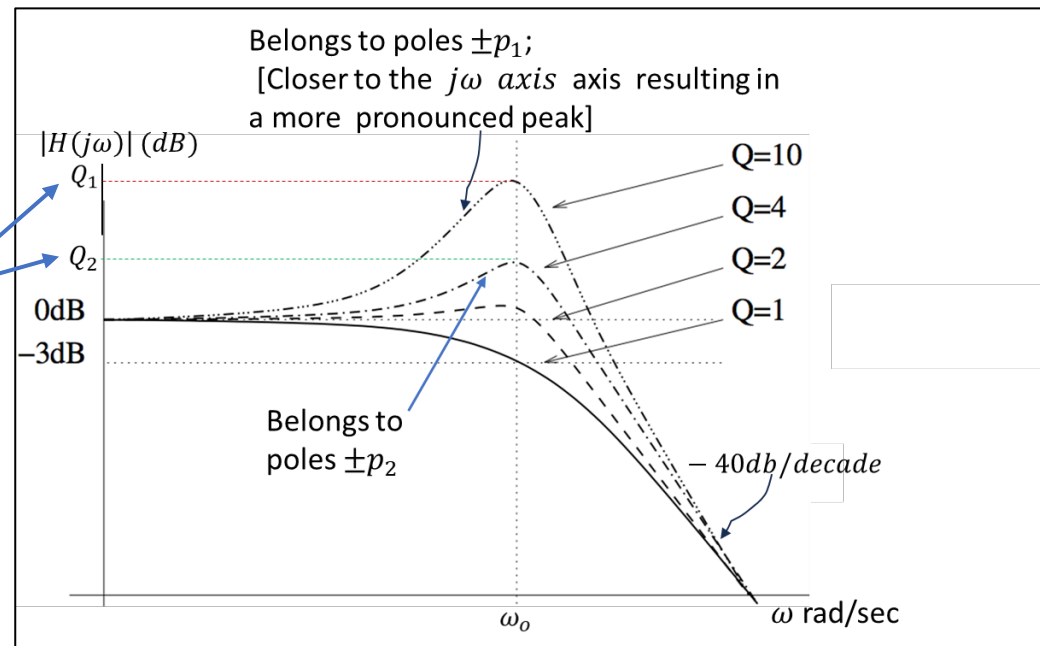


Keeping ω_0 constant and varying Q is equivalent to moving the poles from the $j\omega$ axis toward the real axis.

$$H(j\omega) = \frac{\omega_o^2}{(j\omega)^2 + \frac{\omega_o}{Q} (j\omega) + \omega_o^2}$$

$$|H(j\omega)| = \frac{\omega_o^2}{\sqrt{(\omega_o^2 - \omega^2)^2 + \left(\frac{\omega\omega_o}{Q}\right)^2}}$$

When $\omega = \omega_o$; $|H(j\omega)| = Q$



2nd order band-pass filter

2nd order band-pass filter

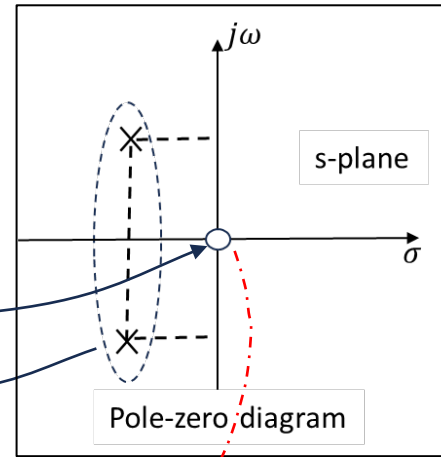
$$H(s) = \frac{b_o s}{s^2 + b_o s + \omega_o^2}$$

This term is not present for low-pass resonant filter

b_o (bandwidth) = $\frac{\omega_o}{Q}$; Q is the **quality factor**; ω_o is the resonant frequency;

$$H(s) = \frac{\frac{\omega_o}{Q} s}{s + \frac{\omega_o}{Q} s + \omega_o^2}$$

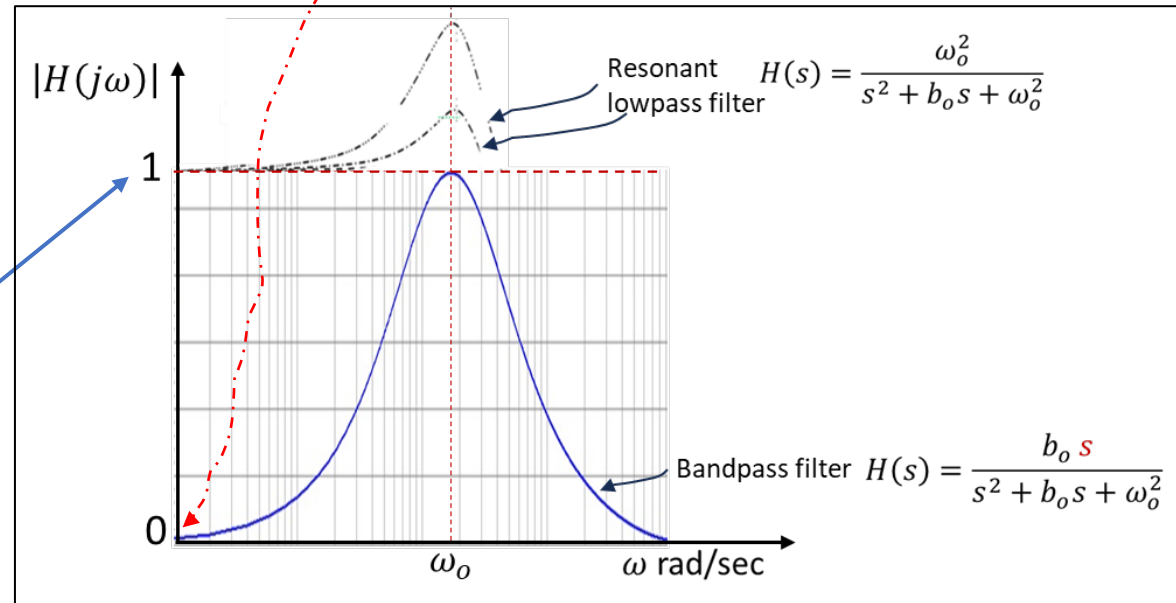
Complex poles



$$H(j\omega) = \frac{\frac{\omega_o}{Q} (j\omega)}{(j\omega)^2 + \frac{\omega_o}{Q} (j\omega) + \omega_o^2}$$

$$|H(j\omega)| = \frac{\frac{\omega_o}{Q} (\omega)}{\sqrt{(\omega_o^2 - \omega^2)^2 + \left(\frac{\omega\omega_o}{Q}\right)^2}}$$

When $\omega = \omega_o$; $|H(j\omega)| = 1$



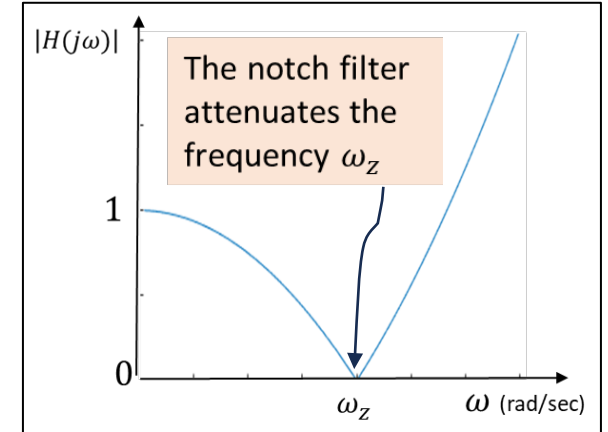
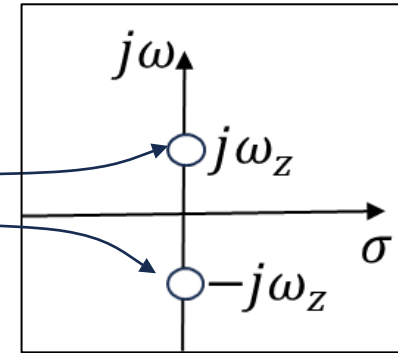
Notch filter and band-stop filter

Notch filter

- A notch filter suppresses a specific frequency (called the notch frequency), ideally reducing its magnitude to zero.

$$H(s) = \frac{s^2 + \omega_z^2}{\omega_z^2} \quad \leftarrow \text{A complex zero located on the } j\omega \text{ axis.}$$

$$H(j\omega) = \frac{(j\omega)^2 + \omega_z^2}{\omega_z^2} ; \quad |H(j\omega)| = \frac{\omega_z^2 - \omega^2}{\omega_z^2} ;$$



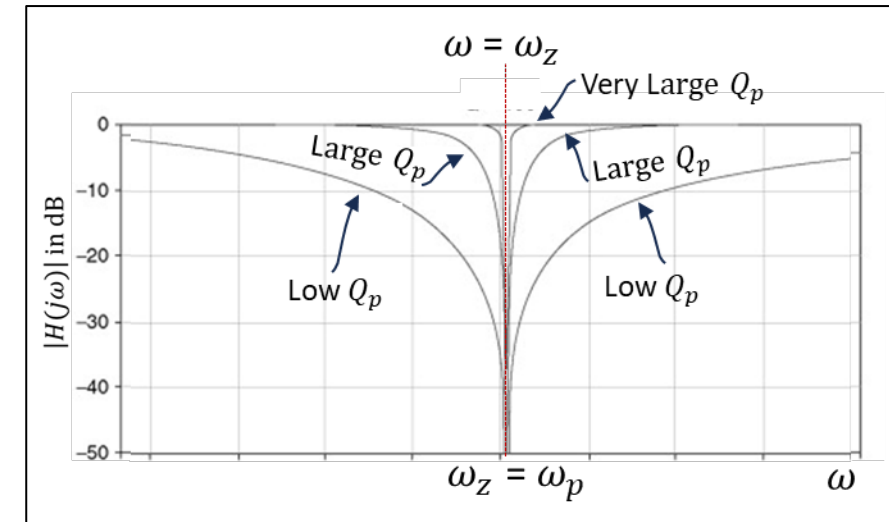
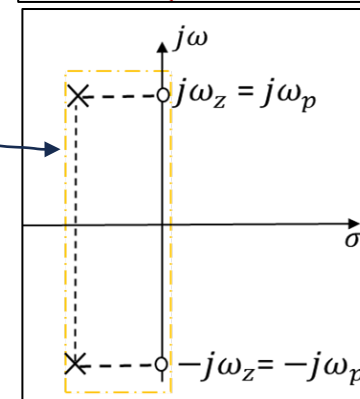
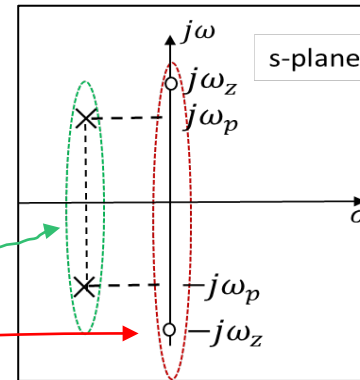
Band-stop filter

- A notch filter becomes a band-stop filter when complex poles are introduced to control the bandwidth around the notch frequency.

$$H(s) = \frac{s^2 + \omega_z^2}{s^2 + b_p s + \omega_p^2}$$

b_p - bandwidth of the resonant pole ; ω_p - resonant frequency of the pole
 Q_p - quality factor of the pole ; ω_z - notch frequency

- When the pole frequency $\omega_p = \omega_z$, we get a standard notch filter, depending on $b_p = \omega_p / Q_p$, the notch will be narrower or wider.



$$H(s) = \frac{s^2 + \omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

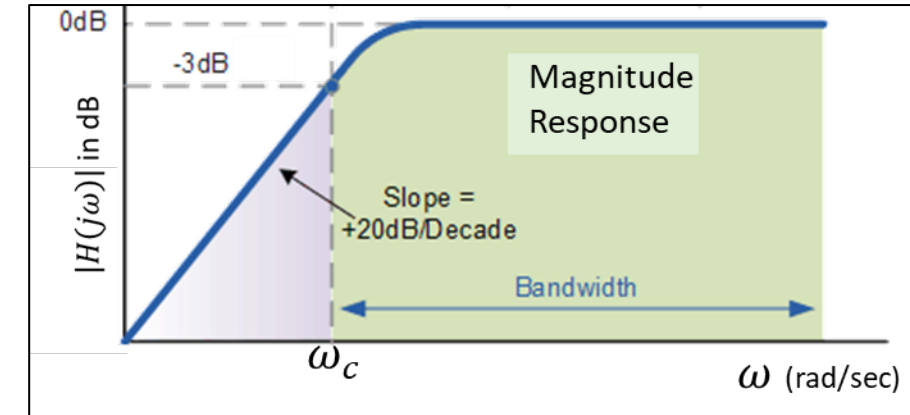
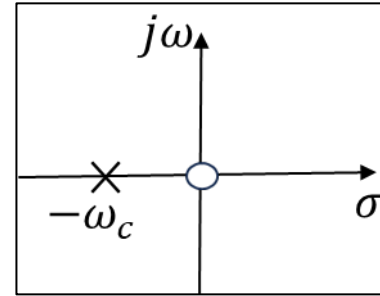
1st order high-pass filter

1st order high-pass filter

- It passes high frequency signals and attenuates low frequency signals.

$$H(s) = \frac{s}{s + \omega_c}; H(j\omega) = \frac{j\omega}{j\omega + \omega_c};$$

- High-pass filter blocks DC, because $|H(j0)| = 0$, but passes high frequencies; $|H(j\omega)| \rightarrow 1$ as $\omega \rightarrow \infty$



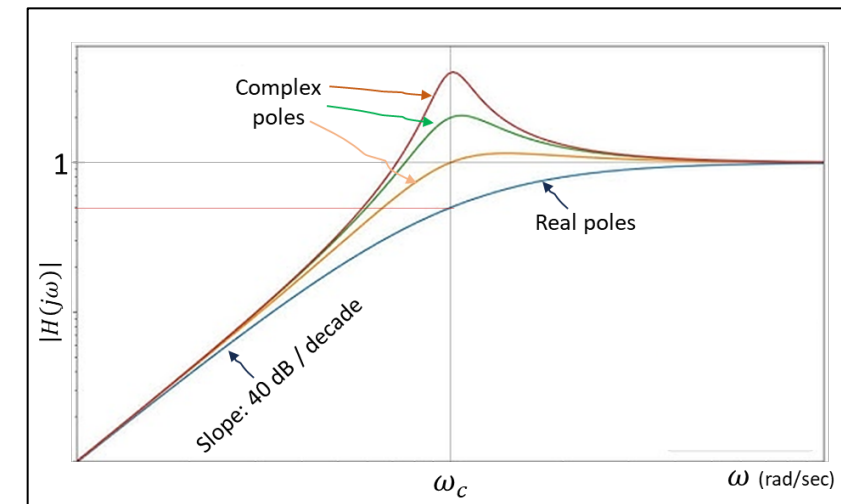
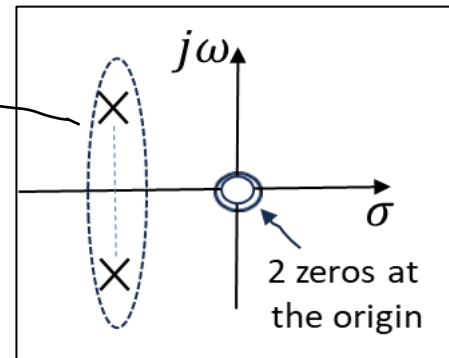
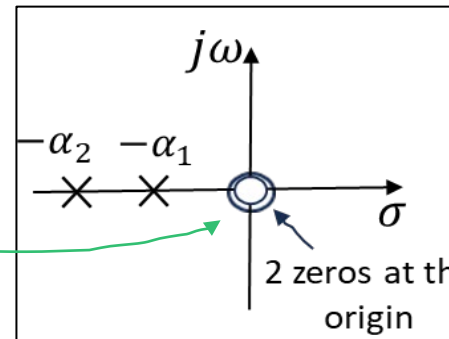
2nd order high-pass filter

- Second order high-pass filter with **real poles**

$$H(s) = \frac{s^2}{s^2 + b_1s + \omega_1^2}$$

- Second order high-pass filter with **complex poles**

$$H(s) = \frac{s^2}{s^2 + b_2s + \omega_2^2}$$



Key Take-Aways

What can poles and zeros tell us?

Feature	Physical meaning
$\Re\{p\}$	decay/growth
$\Im\{p\}$	oscillation frequency
pole near $j\omega$ -axis	strong/slow-decaying response
zero near $j\omega$	attenuation near that frequency

Time response vs frequency response

$H(s)$ splits into two viewpoints:

Time-response view: Poles $\rightarrow$ transient/natural response

Frequency-response view: Evaluate $H(j\omega)$ $\rightarrow$ magnitude + phase vs frequency

This is the conceptual bridge from Week 4 to Week 5.

Signal spectrum vs system response

$X(\omega)$ = what frequencies are in the input

$H(j\omega)$ = how the system treats each frequency

$$Y(\omega) = H(j\omega)X(\omega).$$

Signal spectrum tells us what is present.

System frequency response tells us what gets passed, attenuated, or phase shifted.

ELEC2134: Circuits and Signals

System Analysis

Tutorial questions

Pass-Level Tutorial Questions (System Analysis)

Question 1: A system has a pair of complex conjugate poles $p_1, p_2 = -1 \pm j2$, a single zero $z_1 = -4$ and a gain $K = 3$. Show that the differential equation representing the system is given by $\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + 5y(t) = 12x(t) + 3\frac{dx(t)}{dt}$

where $x(t)$ is the input and $y(t)$ is the output of the system.

Question 2: A system has a transfer function $H(s) = \frac{3s-1}{(s+3)(s-2)}$. Show that the impulse response of the system is given by:

$$h(t) = 2e^{-3t}u(t) + e^{2t}u(t), \text{ if the system is causal. Can the system be both stable and causal?}$$

Question 3: Sketch the magnitude responses for the systems described by the following transfer functions using the relationship between the pole and zero locations and the $j\omega$ axis in the s-plane:

$$(a) H(s) = \frac{s}{s^2+2s+101}; \quad (b) H(s) = \frac{s^2+25}{s+1}; \quad (c) H(s) = \frac{s-2}{s+2}$$

Question 4: Sketch the magnitude response of the system having the transfer function $H(s) = \frac{s-0.5}{(s+0.1-5j)(s+0.1+5j)}$ and Find the values of $|H(j0)|$ and $|H(j5)|$.

Question 5:

(a) Show that the transfer function of (i) an ideal delay of T sec is e^{-sT} ; (ii) an ideal differentiator is s ; (iii) an ideal integrator $1/s$.

(b) Find and sketch the frequency response (magnitude and phase response) for the three systems given above in (a).

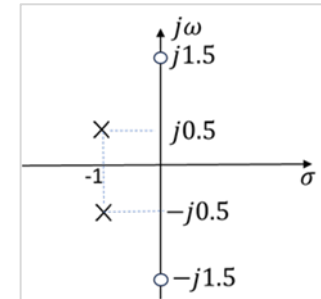
Question 6: Find the cascade and parallel realisation of $H(s) = \frac{s+3}{s^2+7s+10}$ and draw the structure for each realisation.

Question 7:

The pole-zero plot of a second order system $H(s)$ is shown on the right.

The dc response of this system is -1 $\{H(j0) = -1\}$.

If $H(s) = k(s^2 + b_1s + b_2)/(s^2 + a_1s + a_2)$, determine the constants, k, b_1, b_2, a_1, a_2 .



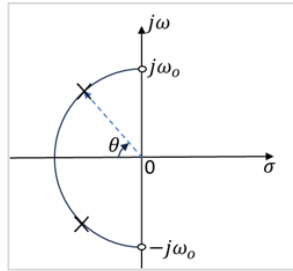
Higher-Level Tutorial Questions (System Analysis)

Question 1: Sketch the phase responses for the systems described by the following transfer functions using the relationship between the pole and zero locations and the $j\omega$ axis in the s-plane.

$$(i) H(s) = \frac{s-1}{s+2}; \quad (ii) H(s) = \frac{s+1}{s+2}; \quad (iii) H(s) = s^2$$

Question 2: The transfer function of a second order notch filter is given by

$$H(s) = \frac{s^2 + \omega_o^2}{s^2 + (2\omega_o \cos(\theta))s + \omega_o^2}$$



and the corresponding pole-zero configuration is shown above. Using the above transfer function, design a second-order notch filter to suppress 50Hz hum in a radio receiver. Sketch the magnitude response $|H(j\omega)|$ for $\theta = 60^\circ$.

Question 3: Design a second-order bandpass filter with center frequency $\omega = 10$. The gain should be zero at $\omega = 0$ and at $\omega = \infty$. Select poles at $-a \pm j10$. Leave your answer in terms of a . Explain the influence of ' a ' on the frequency response.

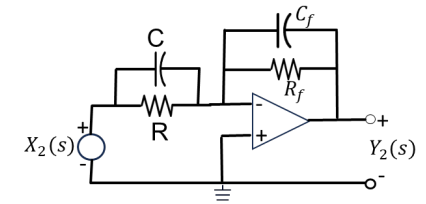
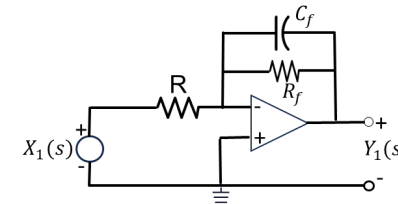
Question 4: Find and sketch the magnitude and phase of the filter whose transfer function is given by

$$H(s) = \frac{s^2 + 2s + 17}{s^2 + 4s + 104}$$

Question 5: Show that the transfer functions of the op-amp circuits shown below are $H_1(s)$ and $H_2(s)$ respectively, where:

$$H_1(s) = \frac{Y_1(s)}{X_1(s)} = \frac{-R_f}{R} \left(\frac{a}{s+a} \right) \quad a = \frac{1}{R_f C_f}$$

$$H_2(s) = \frac{Y_2(s)}{X_2(s)} = \frac{-C}{C_f} \left(\frac{s+b}{s+a} \right) \quad a = \frac{1}{R_f C_f} \text{ and } b = \frac{1}{RC}$$



Question 6: A transfer function of a bandpass filter is given below:

$$H(s) = \frac{b_p s}{s^2 + b_p s + \omega_p^2}$$

where ω_p is the resonant frequency, b_p is the 3dB bandwidth and Q is the quality factor and is given by $Q = \frac{\omega_p}{b_p}$

Show that the magnitude response of the filter is as given by

$$|H(\omega)| = \frac{\omega \frac{\omega_p}{Q}}{\sqrt{(\omega_p^2 - \omega^2)^2 + \left(\frac{\omega \omega_p}{Q}\right)^2}}$$

and show that the magnitude response at $\omega = \omega_p$ is unity.